Mermaid Flowchart Customization Guide

Deep dive into flowchart customization — shapes, colors, layout, styling, and advanced patterns.

Node Shapes

Every shape has semantic meaning. Choose based on the node's purpose.

[Diagram 1 failed to render]

Shape Syntax Quick Reference:

Shape	Syntax	Use Case
Rectangle	`[text]`	Standard process
Rounded	`([text])`	Start/end points
Stadium	`([text])`	Same as rounded
Subroutine	`[[text]]`	Function call
Cylindrical	`[(text)]`	Database
Circle	`((text))`	Connection point
Double Circle	`(((text)))`	-
Asymmetric	`>text]`	Choice/flag
Rhombus	`{text}`	Decision
Hexagon	`{{text}}`	Preparation
Parallelogram (input)	`[/text/]`	Input data
Parallelogram (output)	`[\text\]`	Output data
Trapezoid	`[/text\]`	Manual operation
Inverted Trapezoid	`[\text/]`	-

Arrow Types and Line Styles

Arrows communicate relationships. Mix types to show different connection meanings.

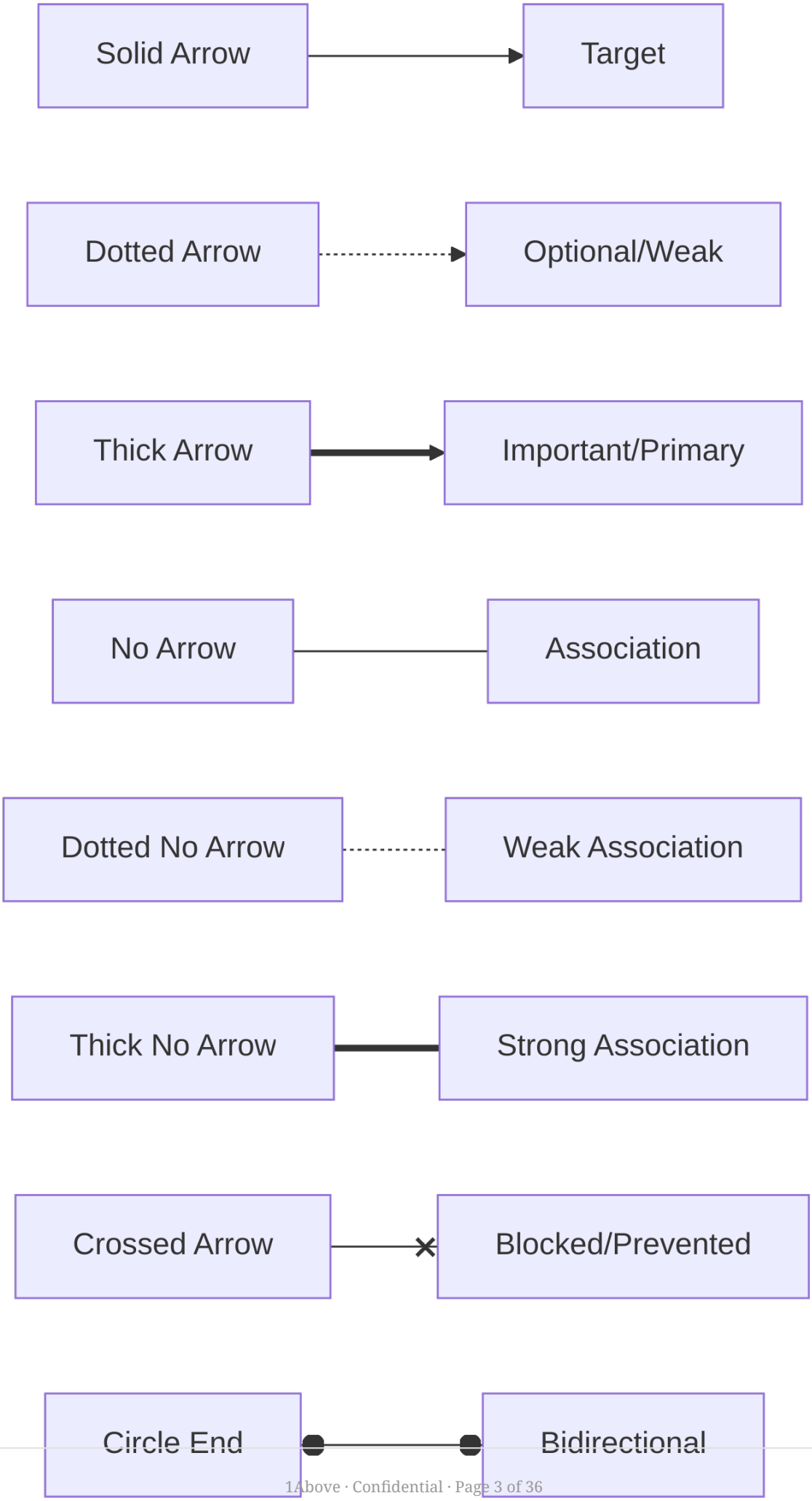
Solid Arrow
Target
Dotted Arrow
Optional/Weak
Thick Arrow
Important/Primary
No Arrow
Association
Dotted No Arrow
Weak Association
Thick No Arrow
Strong Association
Crossed Arrow
Blocked/Prevented
Circle End
Bidirectional

Diagram 2

Arrow Syntax:

Pattern	Code	Meaning
Solid line, arrow	`-->`	Standard flow
Dotted line, arrow	`-.->`	Optional/fallback
Thick line, arrow	`==>`	Primary/critical path
Solid line, no arrow	`---`	Association
Dotted line, no arrow	`-.-`	Weak association
Crossed	`--x`	Blocked
Circle ends	`o--o`	Bidirectional
Double arrow	`<-->`	Two-way communication

Adding Labels:

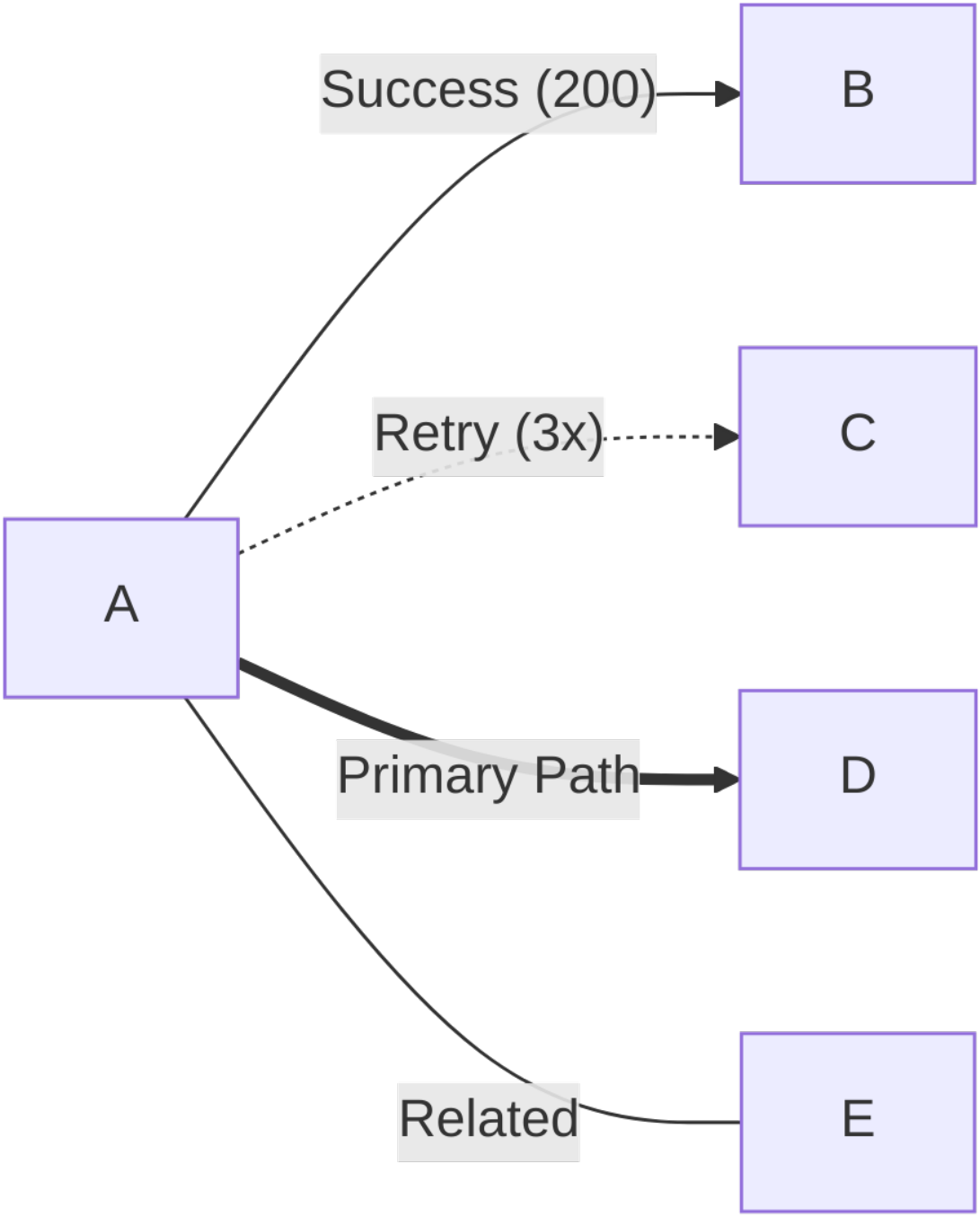

Diagram 3

Longer Arrows (add dashes):

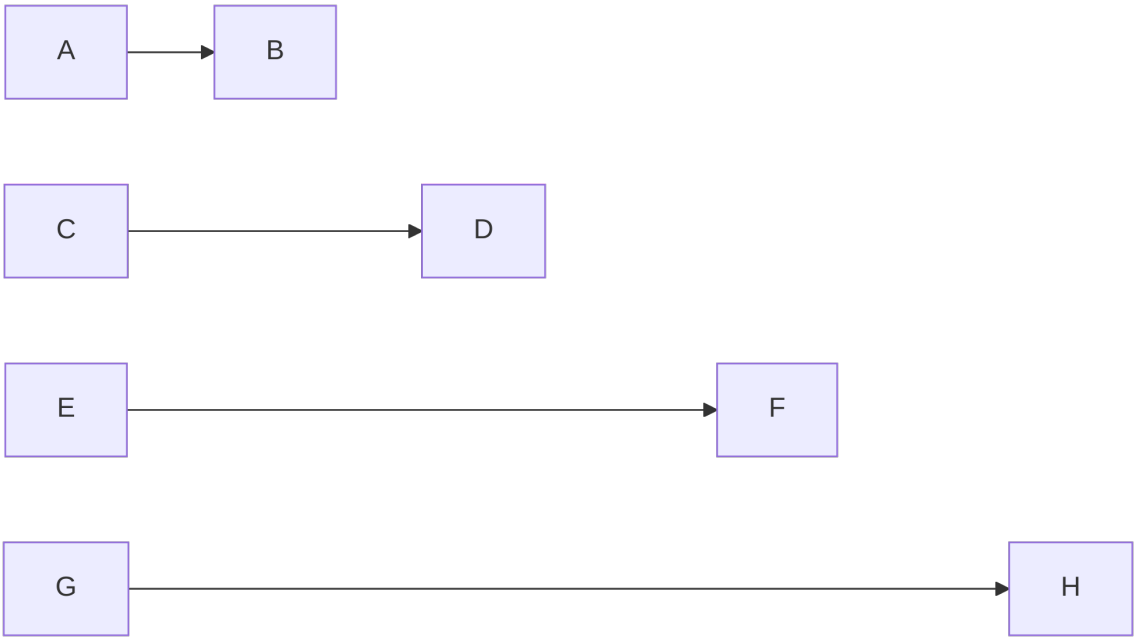

Diagram 4

Color and Styling with classDef

`classDef` creates reusable style classes. Apply to multiple nodes for consistency.

Basic classDef Pattern

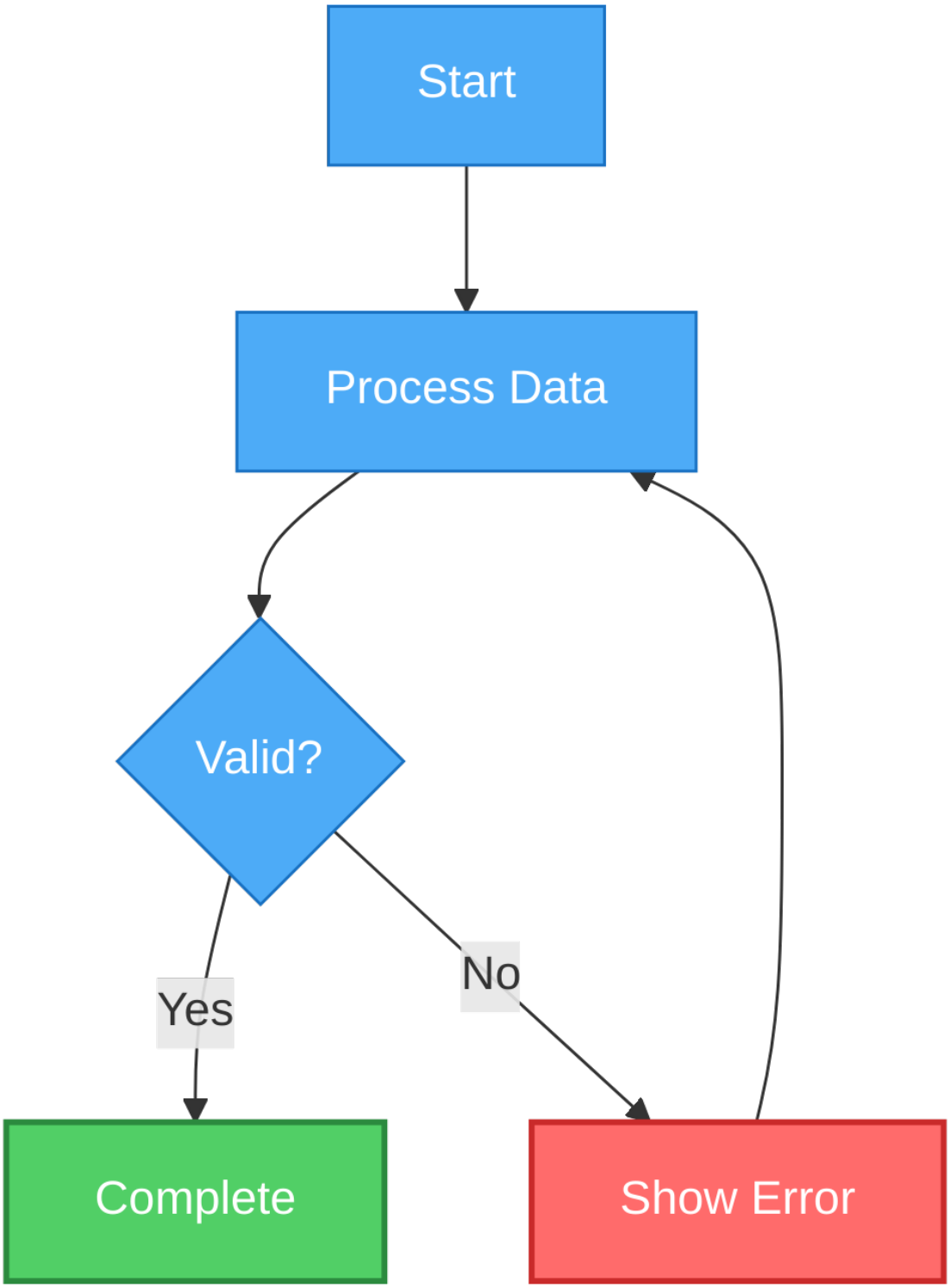

Diagram 5

classDef Syntax:

```
classDef <className> <property>:<value>,<property>:<value>,...
class <nodeId> <className>
```

Common Properties:

Property	Values	Example
`fill`	Color hex/rgb	`fill:#ff6b6b`
`stroke`	Border color	`stroke:#c92a2a`
`stroke-width`	Border thickness	`stroke-width:3px`
`color`	Text color	`color:#fff`
`stroke-dasharray`	Dash pattern	`stroke-dasharray: 5 5`
`rx`	Corner radius	`rx:10px`
`ry`	Corner radius	`ry:10px`

Multiple Classes on One Node

Diagram 6

Inline Styling (Quick One-offs)

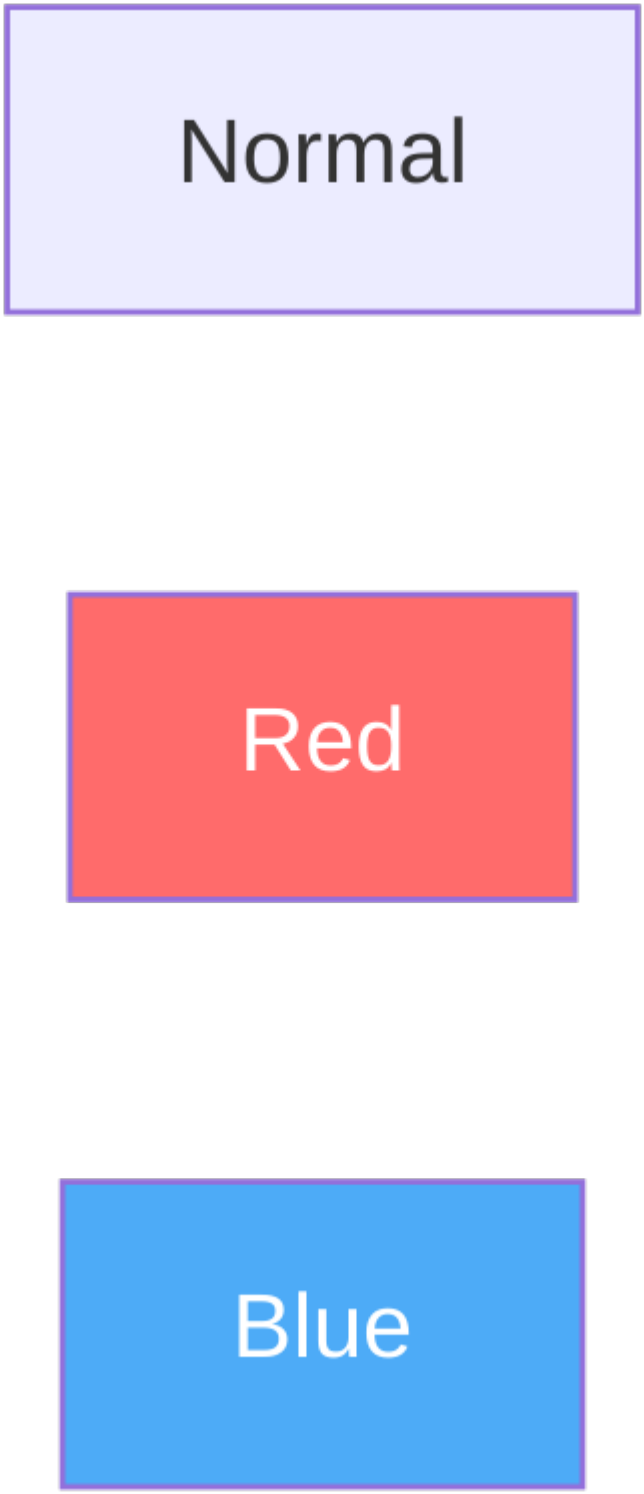

Diagram 7

Theme Customization

Apply global themes to entire diagrams.

Pre-built Themes

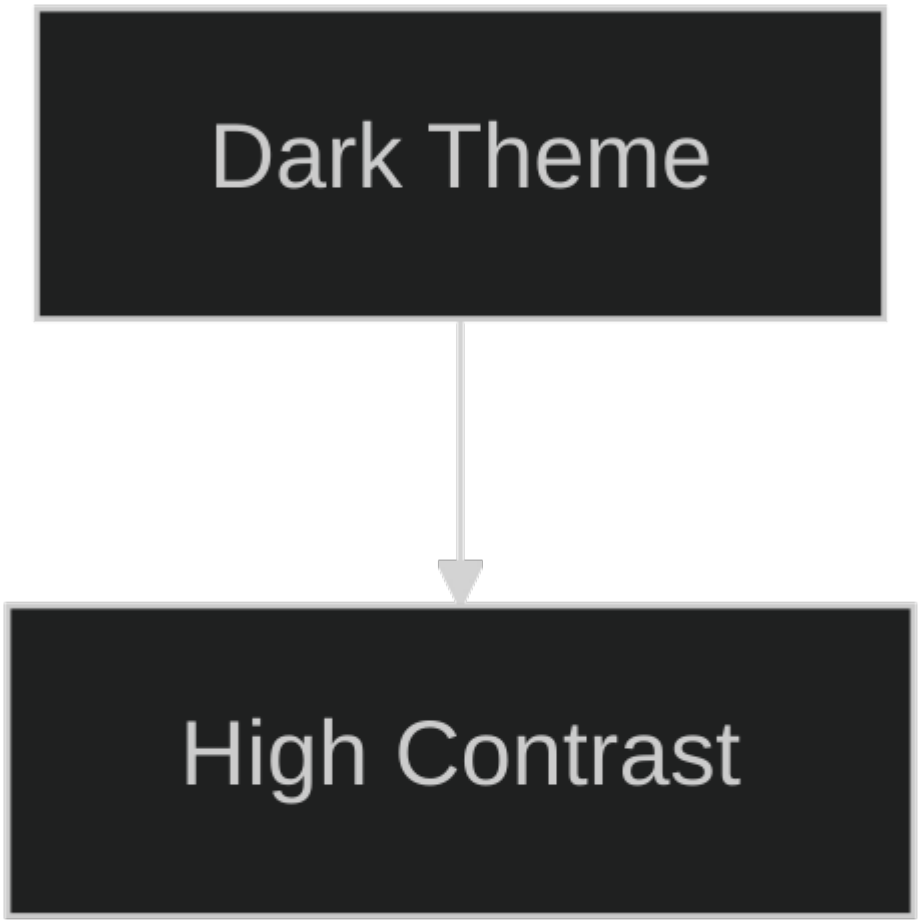

Diagram 8

Available themes: `default`, `dark`, `forest`, `neutral`, `base`

Custom Theme Variables

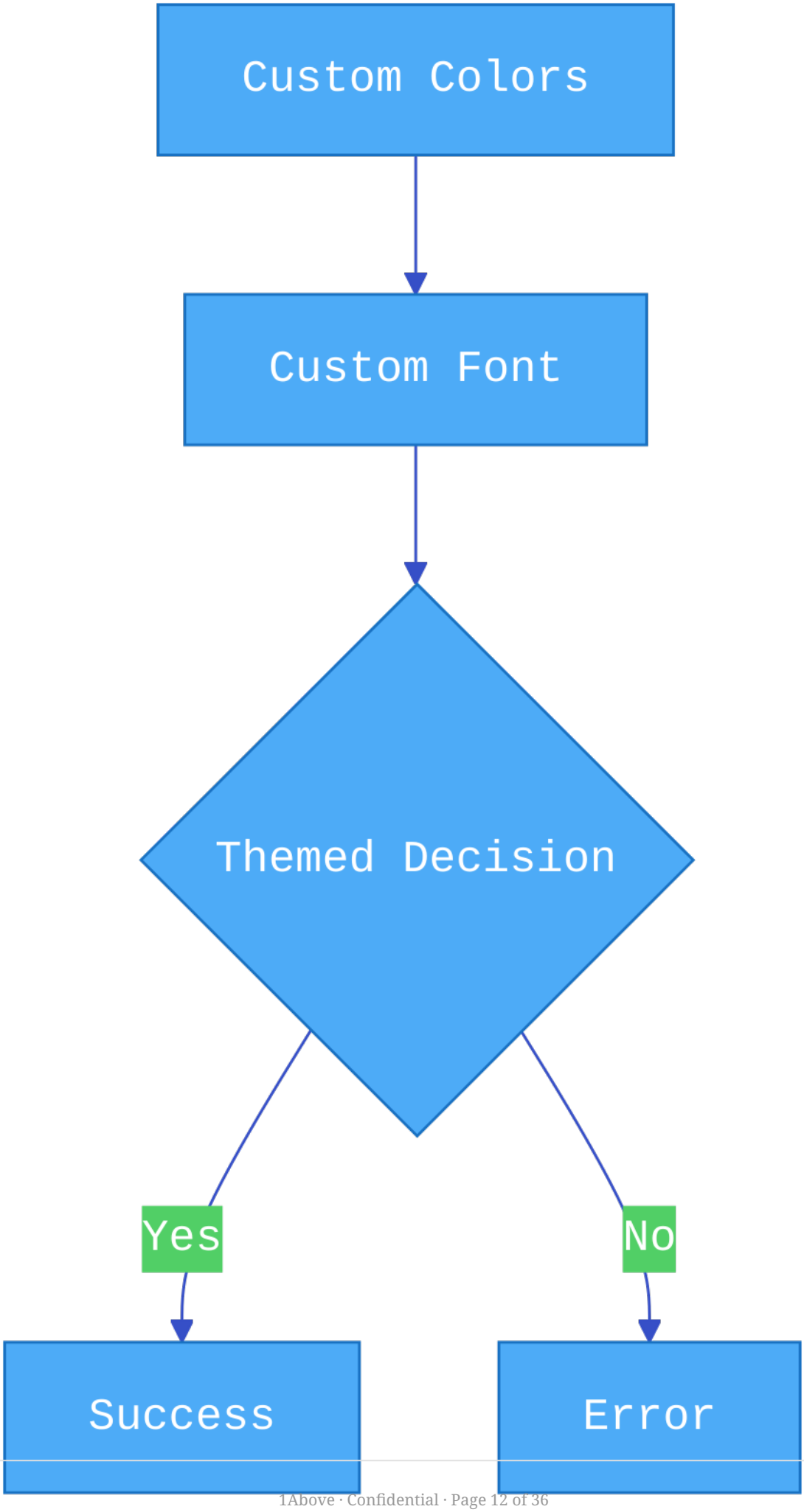
Custom Colors
Custom Font
Themed Decision
Yes
No
Success
Error

Diagram 9

Common Theme Variables:

```
{
  'primaryColor': '#color',        // Main nodes
  'primaryTextColor': '#color',    // Text on primary
  'primaryBorderColor': '#color',  // Border on primary
  'secondaryColor': '#color',      // Secondary nodes
  'tertiaryColor': '#color',       // Tertiary nodes
  'lineColor': '#color',           // Connector lines
  'fontSize': '16px',              // Base font size
  'fontFamily': 'sans-serif',      // Font family
  'background': '#color',          // Diagram background
  'mainBkg': '#color',             // Node background
  'textColor': '#color',           // Default text color
  'edgeLabelBackground': '#color'  // Label background
}
```

Layout and Direction

Control diagram orientation and flow.

Diagram 10

Diagram 11

Diagram 12

Diagram 13

Direction Codes:

- `TD` or `TB` — Top Down / Top to Bottom (default)
- `LR` — Left to Right
- `RL` — Right to Left
- `BT` — Bottom to Top

Subgraphs for Grouping

Subgraphs visually group related nodes and can have their own direction.

Basic Subgraph

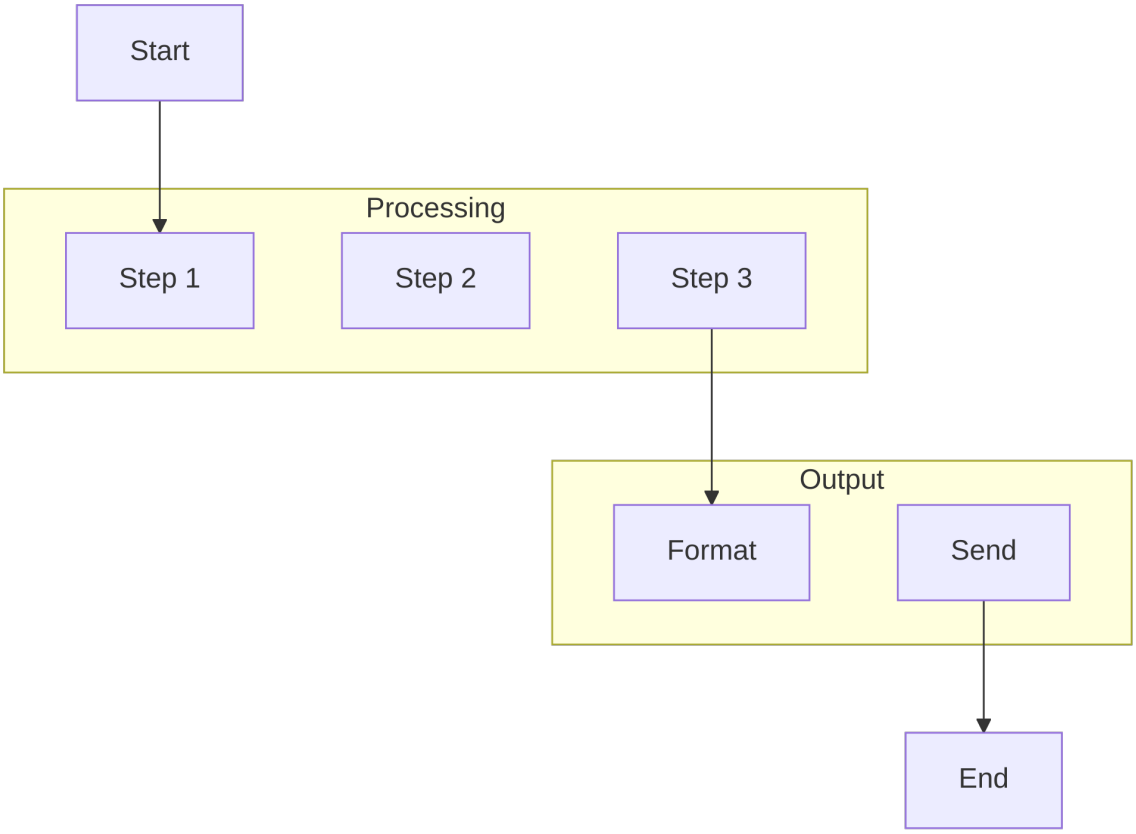

Diagram 14

Named Subgraphs with Custom Direction

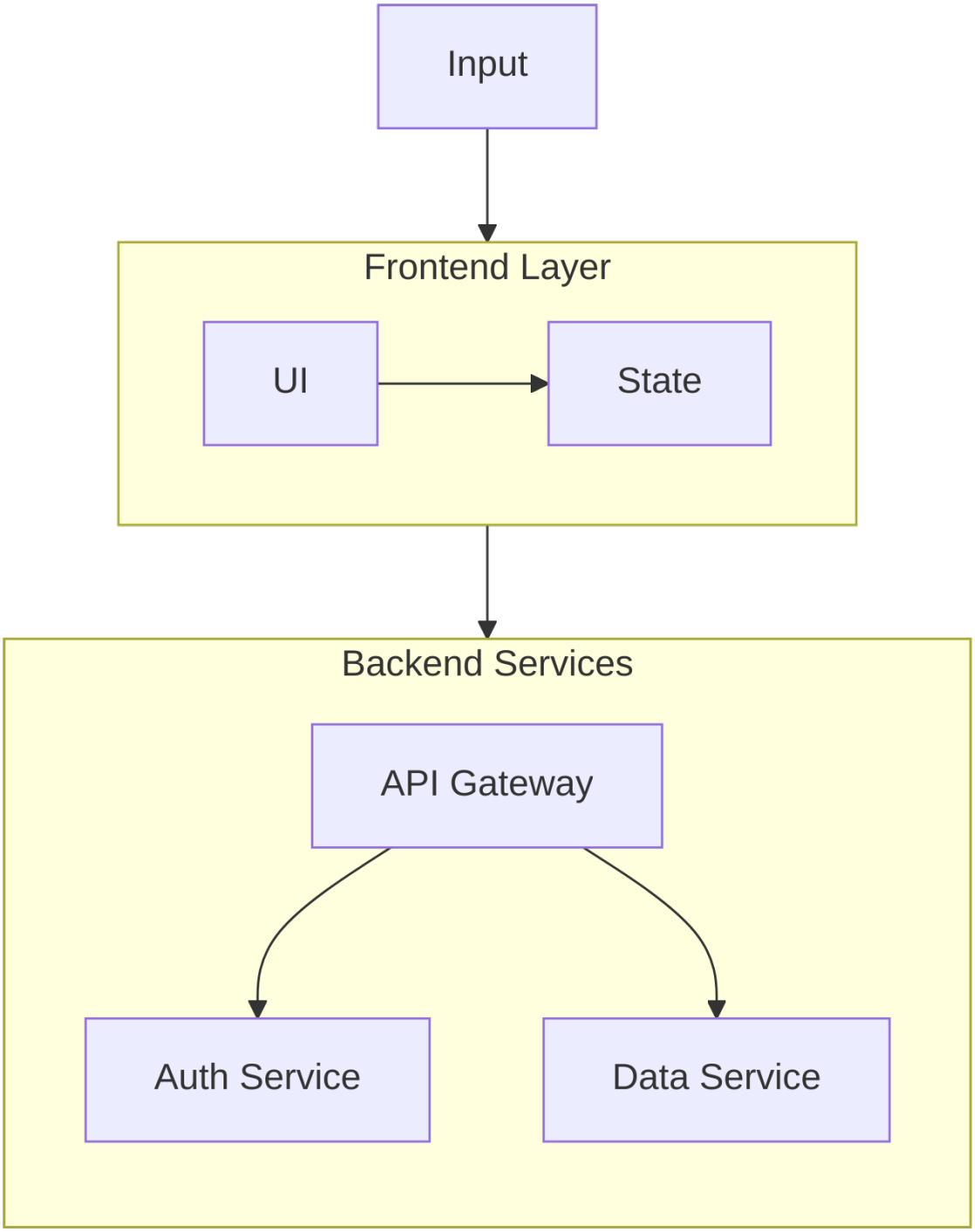

Diagram 15

Nested Subgraphs

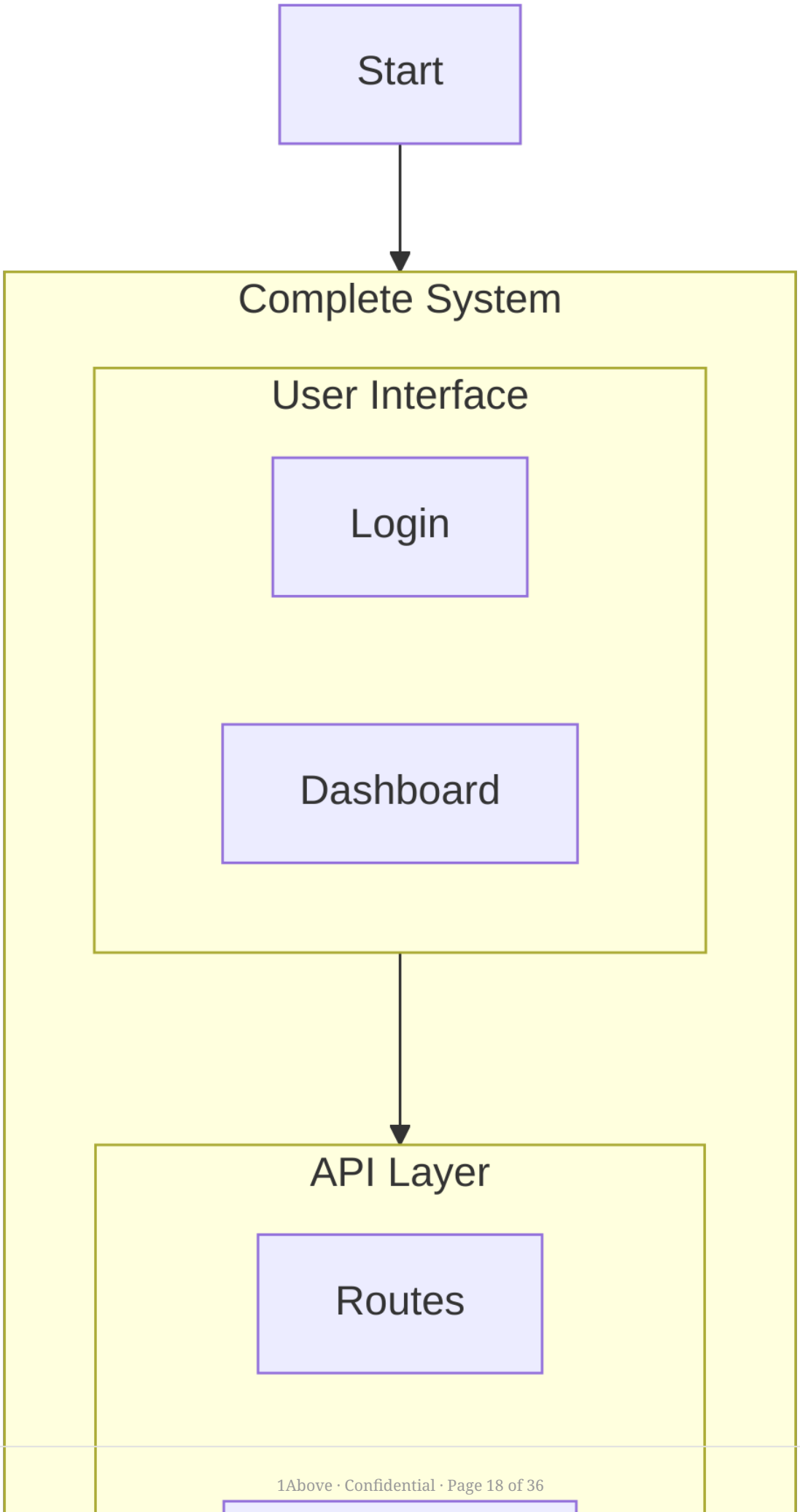
Start
Complete System
User Interface
Login
Dashboard
API Layer
Routes

Diagram 16

Styled Subgraphs

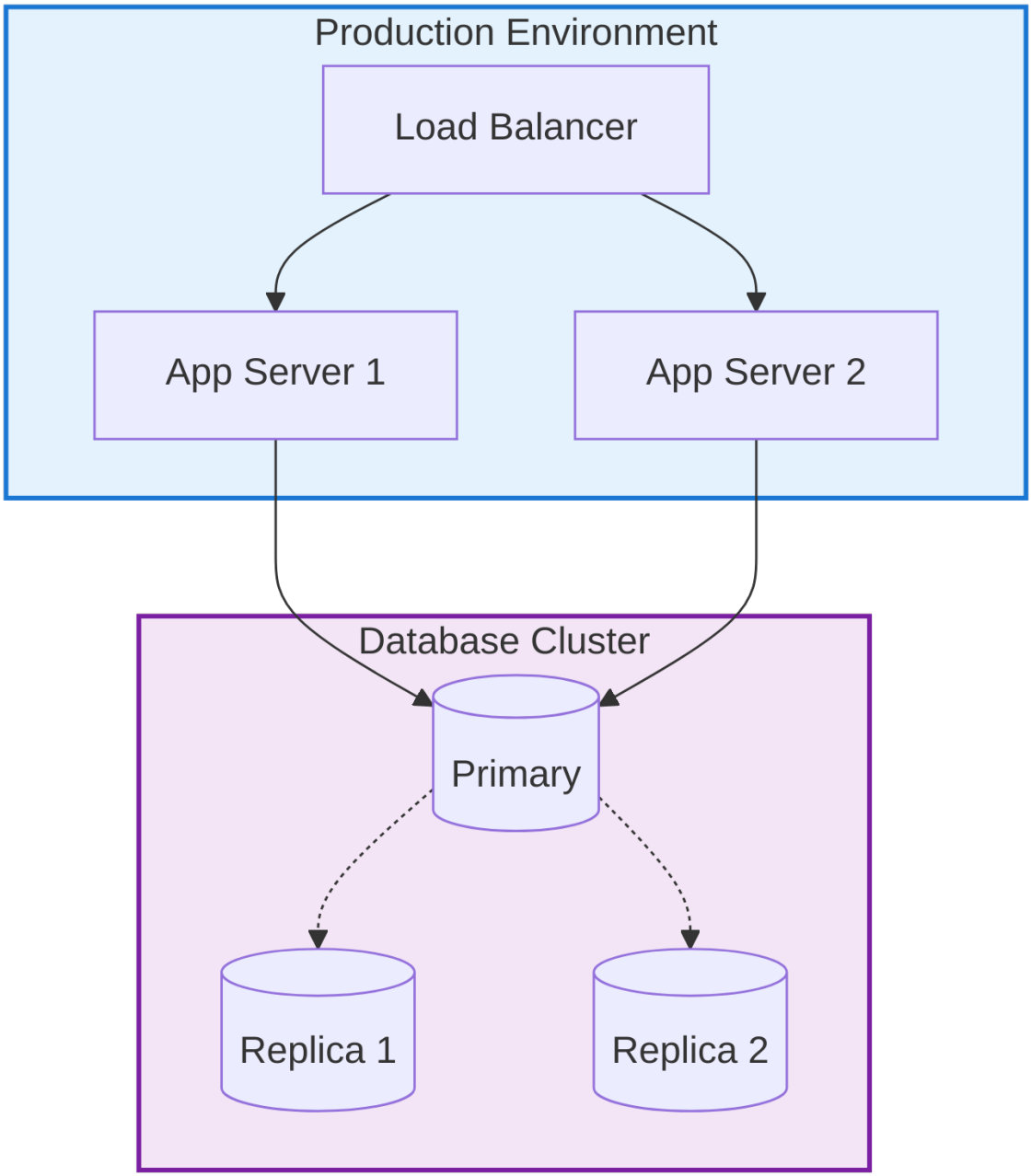

Diagram 17

Advanced Styling Patterns

Individual Node Styling

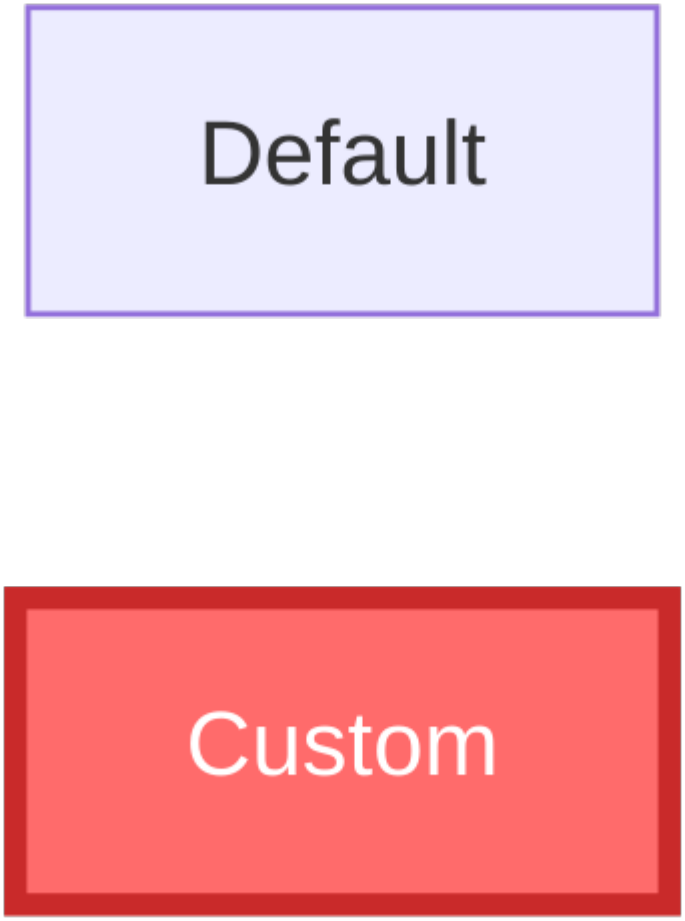

Diagram 18

Link Styling by Index

Diagram 19

Link Styling Properties:

```
linkStyle <index> stroke:<color>,stroke-width:<width>,stroke-dasharray:<pattern>
```

Default Styles for All Nodes/Links

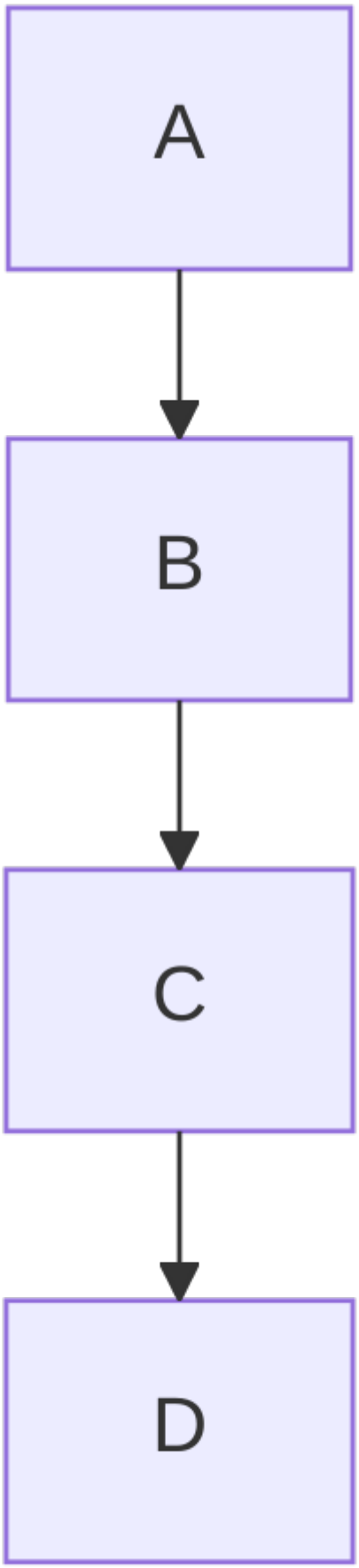
A
B
C
D

Diagram 20

Curve Options:

- `basis` — Smooth curves
- `linear` — Straight lines
- `step` — Step-like connections
- `stepBefore` — Step before
- `stepAfter` — Step after

Practical Styling Patterns

Status Indicators

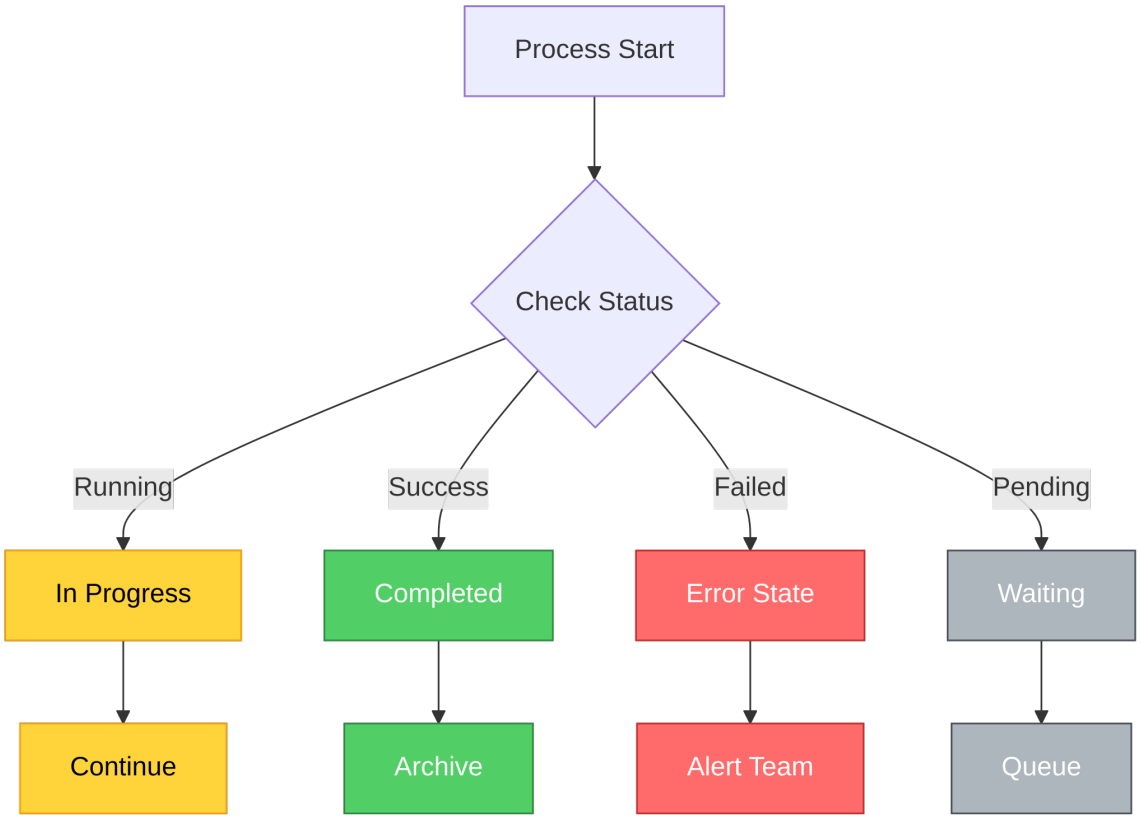

Diagram 21

Priority Levels

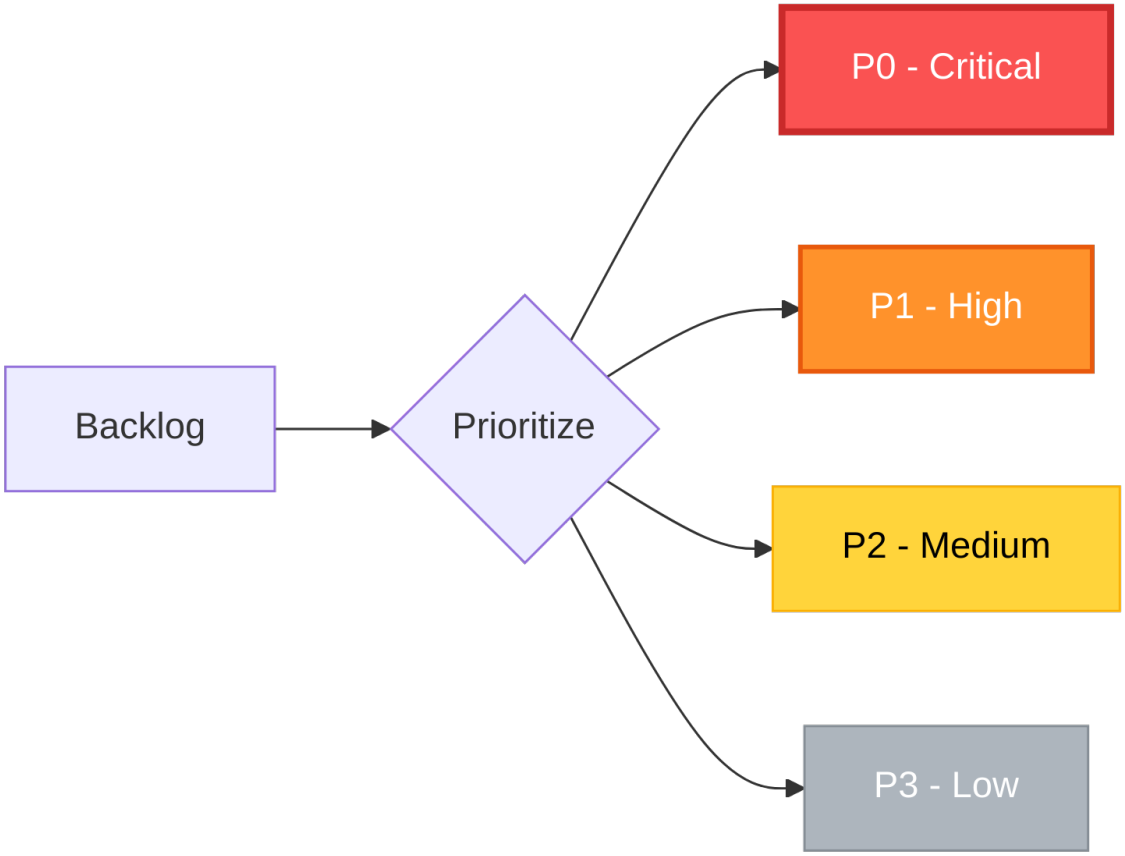

Diagram 22

Environment Layers

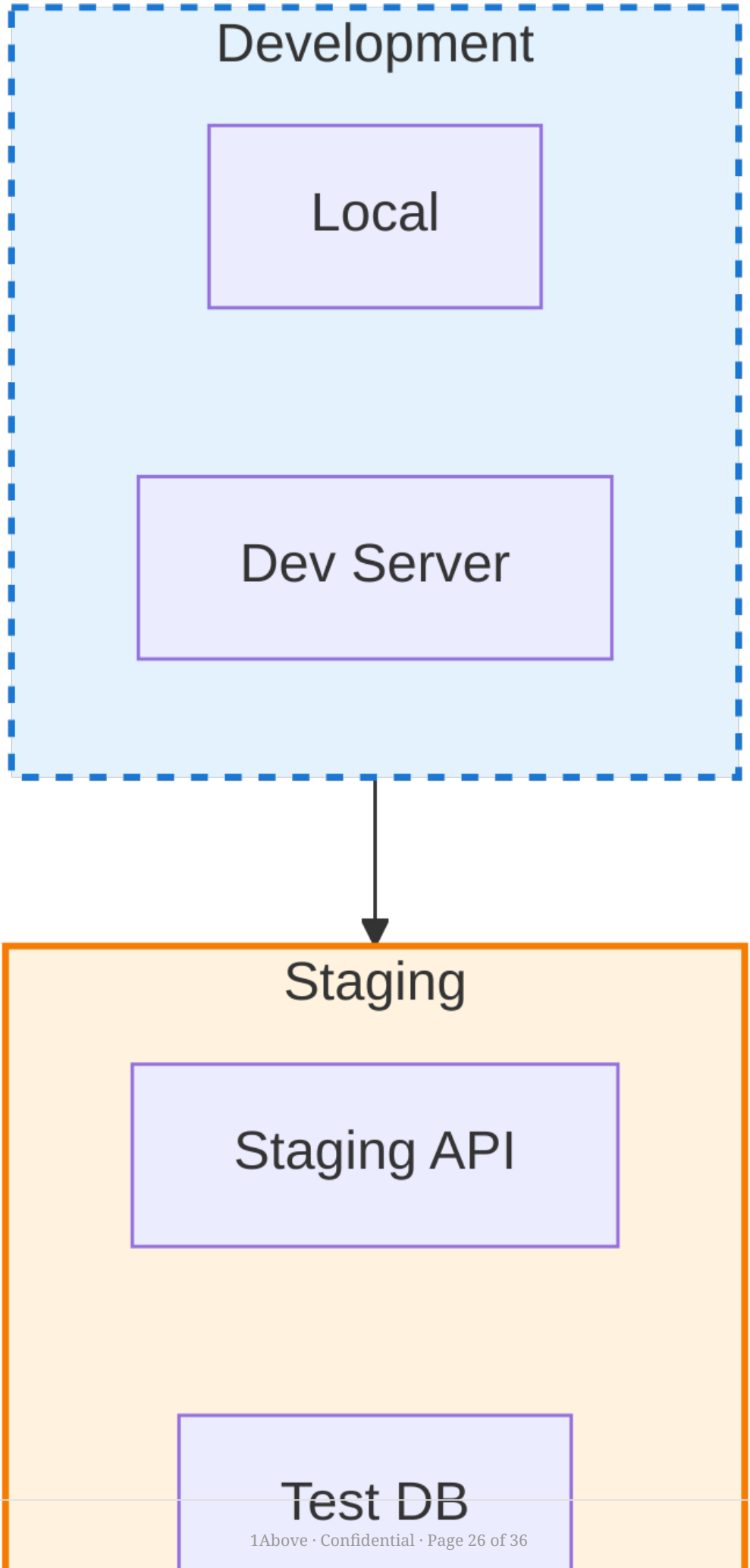
Development
Local
Dev Server
Staging
Staging API
Test DB

Diagram 23

Data Flow with Emphasis

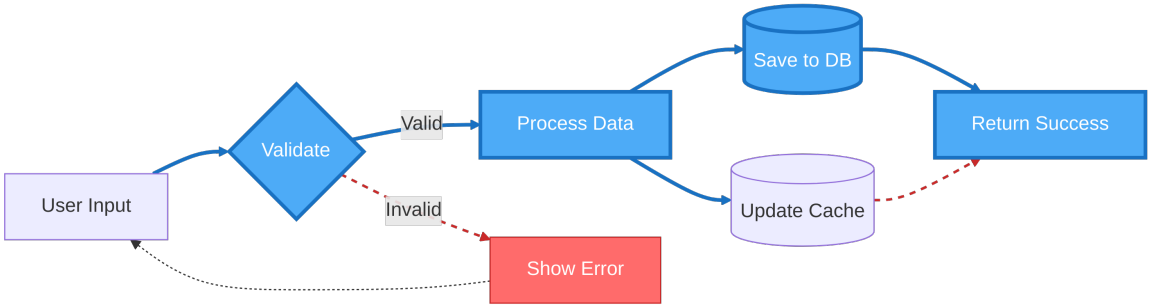

Diagram 24

Interactive Features

Clickable Nodes

[Diagram 25 failed to render]

Click Syntax:

```
click <nodeId> "<url>" "<tooltip>" [_blank/_self/_parent/_top]
click <nodeId> href "<url>" "<tooltip>"
click <nodeId> call <function>() "<tooltip>"
```

Tooltips (via title attribute)

[Diagram 26 failed to render]

Layout Optimization

Managing Complex Layouts

Problem: Messy crossing lines

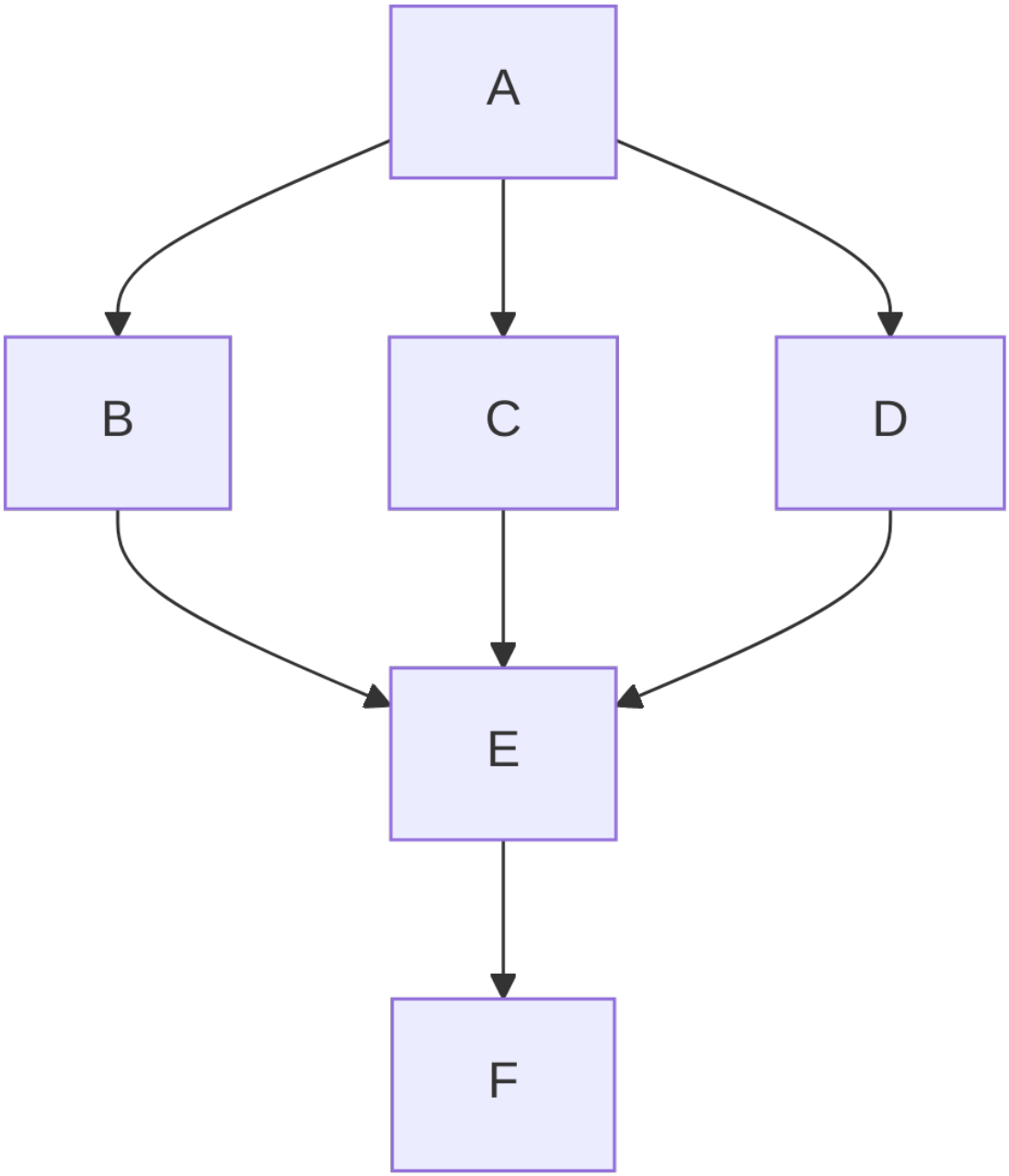

Diagram 27

Solution: Use invisible nodes for routing

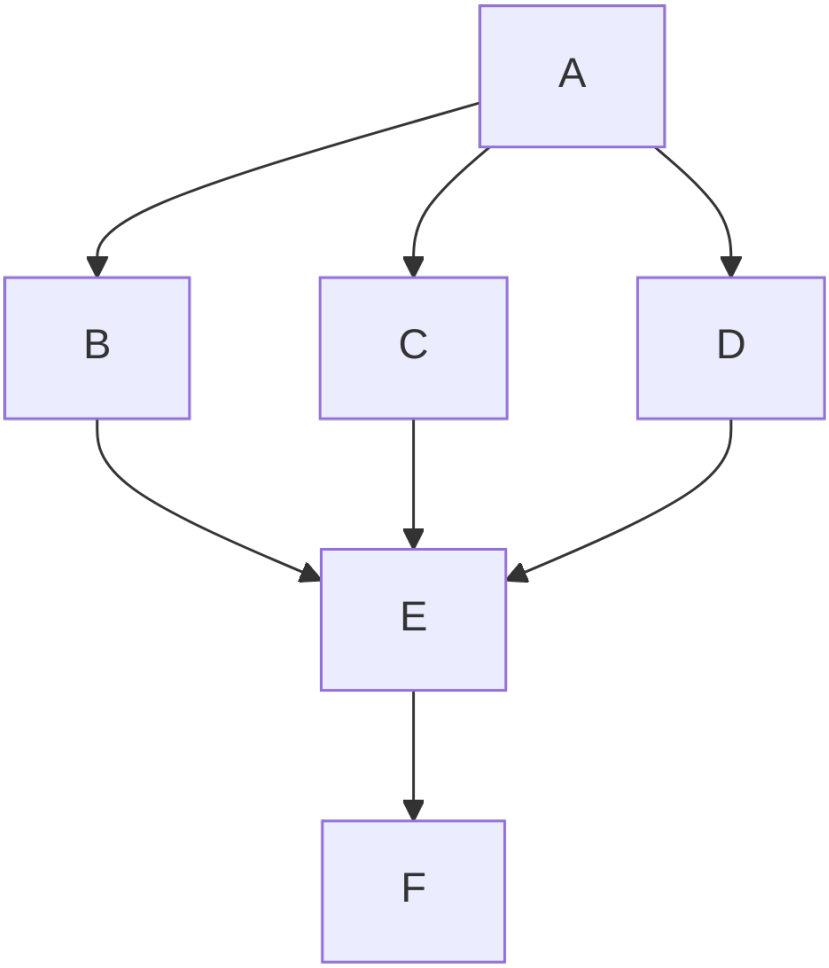

Diagram 28

Rank Separation Control

Force nodes onto the same level:

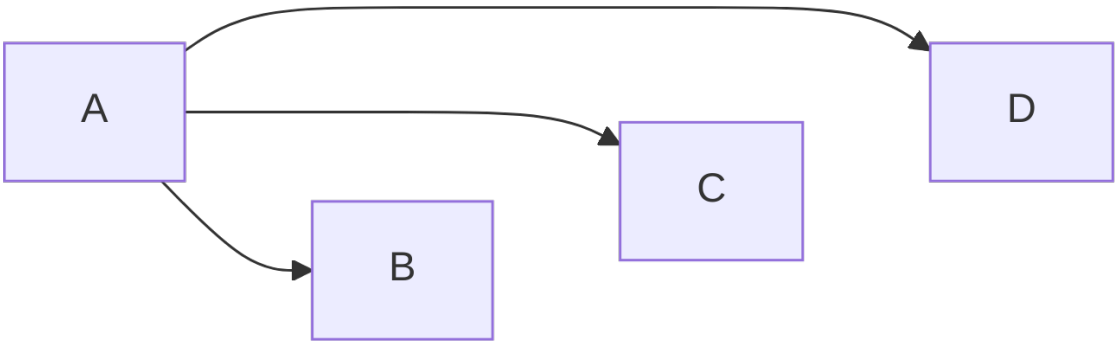

Diagram 29

Real-World Examples

CI/CD Pipeline

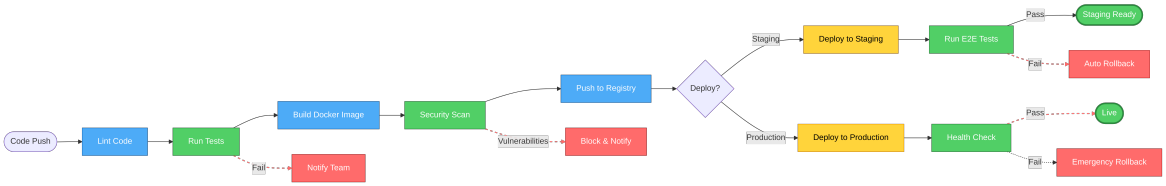

Diagram 30

Authentication Flow with Error Handling

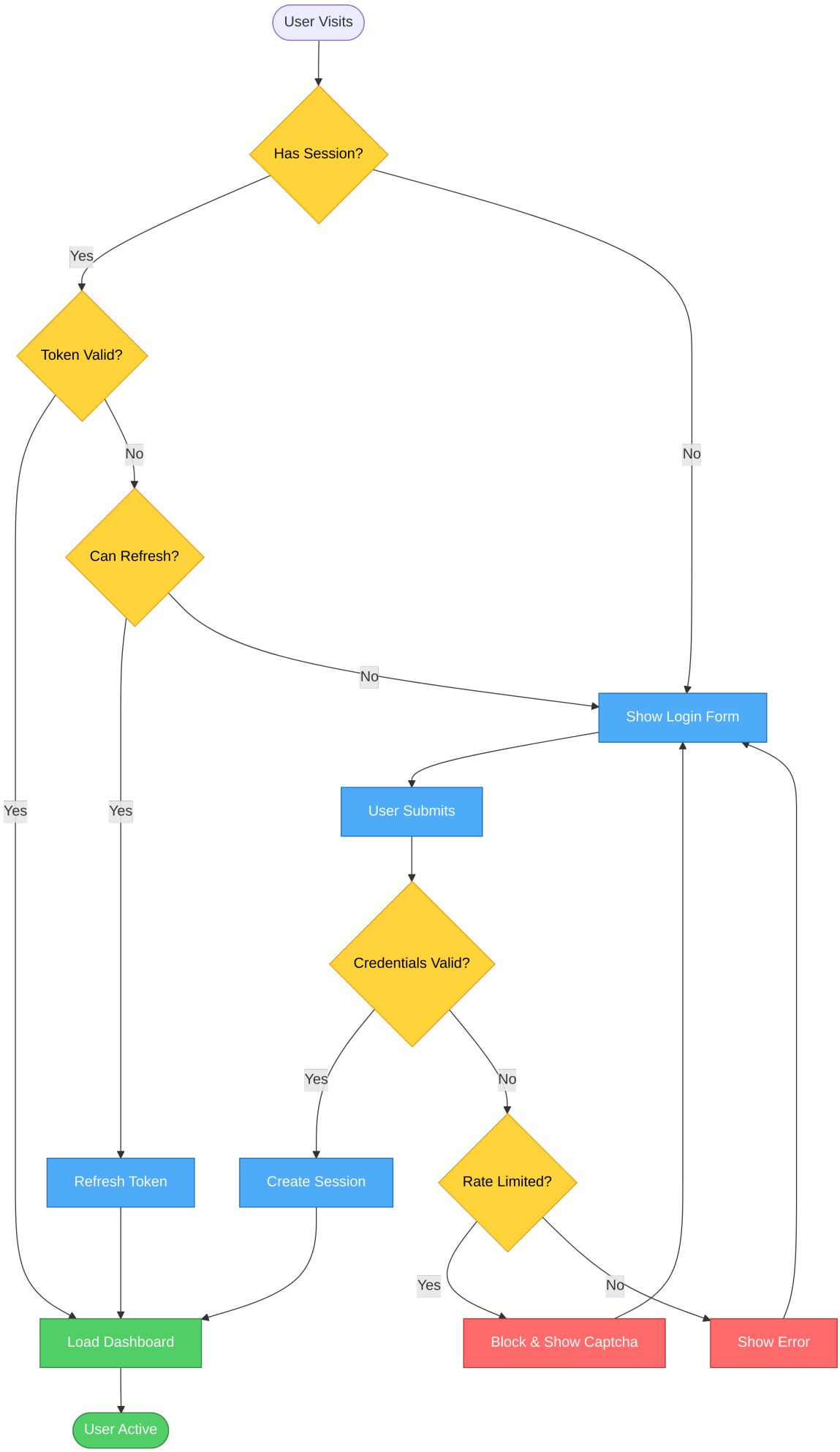
User Visits
Has Session?
Yes
No
Token Valid?
Yes
No
Can Refresh?
Yes
No
Show Login Form
User Submits
Credentials Valid?
Yes
No
Rate Limited?
Yes
No
Refresh Token
Create Session
Load Dashboard
Block & Show Captcha
Show Error
User Active

Diagram 31

Debugging Tips

1. **Test incrementally** — Add nodes/styling one at a time
2. **Use mermaid.live** for real-time preview
3. **Comment out sections** to isolate issues:

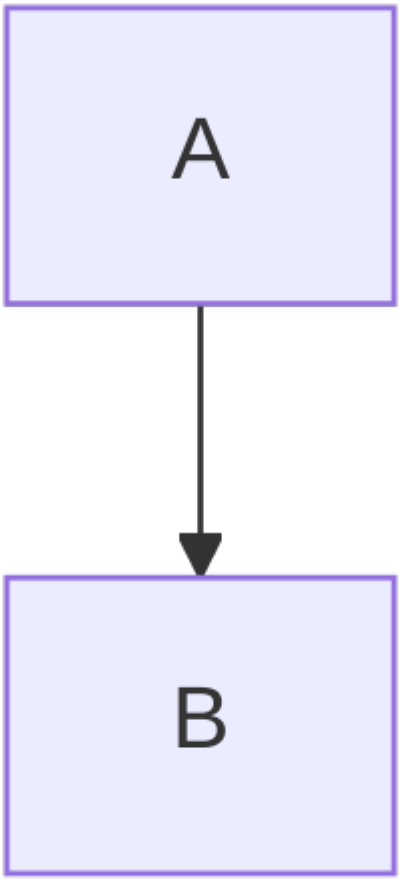

Diagram 32

1. **Validate class names** — must be alphanumeric
2. **Check link indices** — `linkStyle 0` is the first arrow, counting from top

---

Cheat Sheet: Property Values

Colors: - Hex: `#ff6b6b` - RGB: `rgb(255, 107, 107)` - RGBA: `rgba(255, 107, 107, 0.5)` - Named: `red`, `blue`, `green`

Sizes: - Pixels: `2px`, `10px` - Percent: `50%` - Em: `1.5em`

Stroke Patterns: - Solid: `stroke-width:2px` - Dashed: `stroke-dasharray: 5 5` - Dotted: `stroke-dasharray: 1 3` - Long dash: `stroke-dasharray: 10 5`

Font Families: - `monospace` - `sans-serif` - `serif` - `'Inter', system-ui, sans-serif`

Common Color Palettes

Status Colors (Friendly)

```
Success: fill:#51cf66,stroke:#2b8a3e,color:#fff
Warning: fill:#ffd43b,stroke:#f59f00,color:#000
Error: fill:#ff6b6b,stroke:#c92a2a,color:#fff
Info: fill:#4dabf7,stroke:#1971c2,color:#fff
Neutral: fill:#adb5bd,stroke:#495057,color:#fff
```

Priority Levels

```
P0 Critical: fill:#fa5252,stroke:#c92a2a,color:#fff
P1 High: fill:#ff922b,stroke:#e8590c,color:#fff
P2 Medium: fill:#ffd43b,stroke:#fab005,color:#000
P3 Low: fill:#94d82d,stroke:#5c940d,color:#000
```

Environment Tiers

```
Dev: fill:#e3f2fd,stroke:#1976d2,color:#000
Staging: fill:#fff3e0,stroke:#f57c00,color:#000
Production: fill:#ffebee,stroke:#c62828,color:#000
```

Quick Customization Workflow

1. **Build structure** (nodes + arrows)
2. **Define classDef** for common styles
3. **Apply classes** to nodes
4. **Style links** if needed
5. **Add subgraphs** for grouping
6. **Apply theme** if needed
7. **Test and refine**

Template:

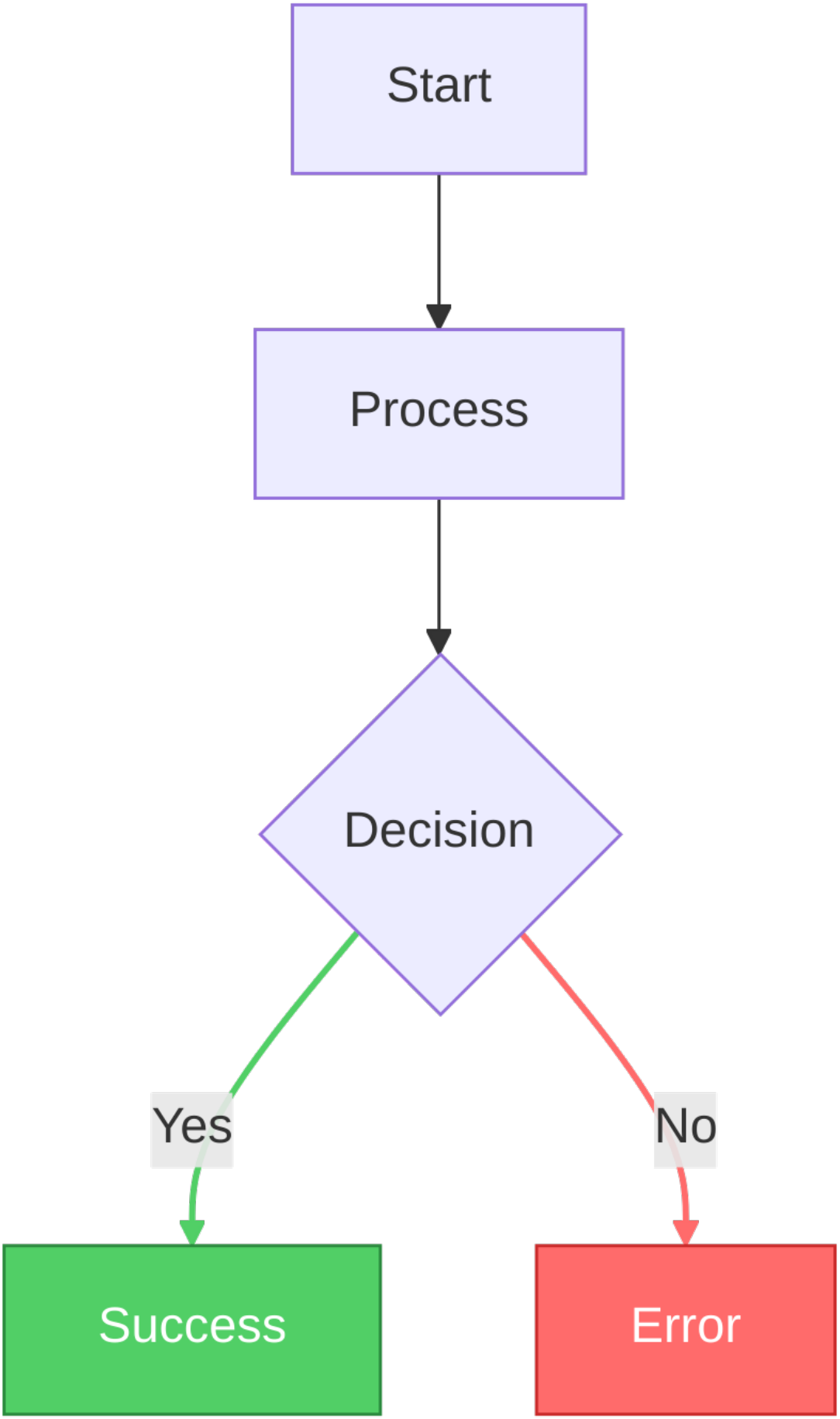
Start
Process
Decision
Yes
No
Success
Error

Diagram 33